

# PPAR Audit

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## Explain why reported portfolio performance changed.

PPAR Audit answers a difficult operational question: "Why did my reported portfolio performance change?" It compares two portfolio-accounting snapshots, quantifies supported causes across holdings, transactions, prices, FX, and related data, flags suspicious source-data relationships, and produces reviewer-ready Excel and HTML reports. When the available evidence is insufficient, the unexplained difference stays visible for human review.

- Everything runs locally, so portfolio data stays inside your environment.
- The Python implementation supports automated batch runs and local customization.
- Standard output includes XLSX, HTML, CSV, JSON, and compact evidence bundles.

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## What PPAR Audit Answers

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PPAR Audit is built around one operational question:

### Why did my reported performance change?

- **Performance Comparison:** identifies changed portfolio and security performance for each time period, quantitatively attributes defensible differences to supported source-data changes, and highlights anything that still needs human review.
- **Data Issues:** flags suspicious source-data relationships — including price ranges, dividend rates, accrued-interest rates, and missing dividends — that may indicate data-quality issues independently of the performance explanation.

# Portfolio Audit Report

## Executive Summary

### PERFORMANCE DIFFERENCES

|                   | Total Quantity | No Performance Differences | Fully Explained Differences | Partly Explained Differences | Unexplained Differences | Setup Incomplete |
|-------------------|----------------|----------------------------|-----------------------------|------------------------------|-------------------------|------------------|
| Portfolios        | 4              | 0                          | 2                           | 1                            | 1                       | 0                |
| Portfolio Periods | 30             | 4                          | 24                          | 1                            | 1                       | 0                |

### DATA ISSUES

| Issue Type                         | Quantity |
|------------------------------------|----------|
| holdings_price_range               | 20       |
| large_price_variation              | 15       |
| duplicate_transactions             | 4        |
| missing_dividend                   | 4        |
| dividend_rate                      | 3        |
| holdings_accrued_rate              | 3        |
| transactions_price_range           | 3        |
| holdings_nonpositive_price         | 2        |
| holdings_stale_price               | 2        |
| pa_sa_rate                         | 2        |
| transaction_security_type_mismatch | 2        |
| transactions_nonpositive_price     | 2        |

## Performance Differences

ROWS: 7

| Portfolio | From Date  | Thru Date  | Performance Difference | Explained Difference | Unexplained Difference | Status           | Comments                                                                                                                                                         |
|-----------|------------|------------|------------------------|----------------------|------------------------|------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ALPHA     | 2026-01-31 | 2026-02-27 | 0.001368               | 0.001368             |                        | Fully Explained  |                                                                                                                                                                  |
| ALPHA     | 2026-05-01 | 2026-05-29 | 0.003571               | 0.003571             |                        | Fully Explained  |                                                                                                                                                                  |
| BALANCED  | 2026-04-01 | 2026-04-10 | -0.002049              | -0.002049            |                        | Fully Explained  |                                                                                                                                                                  |
| BALANCED  | 2026-05-09 | 2026-05-14 | 0.000149               | -0.000051            | 0.000200               | Partly Explained | Possible cause: csus.JPM transactions.amount increased by 200.00 on 2026-05-12. Add YAML configuration to count it as explained.                                 |
| INCOME    | 2026-01-01 | 2026-01-30 | -0.000060              | -0.000060            |                        | Fully Explained  |                                                                                                                                                                  |
| INCOME    | 2026-02-28 | 2026-03-31 | 0.000002               | 0.000002             |                        | Fully Explained  |                                                                                                                                                                  |
| INCOME    | 2026-04-01 | 2026-04-30 | 0.000350               | 0.000000             | 0.000350               | Unexplained      | The Unexplained Difference may be due to missing source-data, source-file timing differences, or vendor methodology that does not match the YAML specifications. |

## Performance Difference Causes

ROWS: 29

| Portfolio | From Date  | Thru Date  | As Of Date | Dataset.Field         | Security      | Snapshot A Value | Snapshot B Value | B - A Difference | Performance Difference Explained | Explanation                                                                                                                                                    |
|-----------|------------|------------|------------|-----------------------|---------------|------------------|------------------|------------------|----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ALPHA     | 2026-01-31 | 2026-02-27 | 2026-01-30 | holdings.accrued      | fius91282Y5Y1 | 4.780000         | 4.830000         | 0.050000         | 0.000000                         | fius91282Y5Y1 beginning holdings.accrued increased by 0.05.                                                                                                    |
| ALPHA     | 2026-01-31 | 2026-02-27 | 2026-01-30 | holdings.market_value | causCASHUSD   | 11495.060000     | 9995.060000      | -1500.000000     | 0.001421                         | causCASHUSD beginning holdings.market_value decreased by 1,500.00.                                                                                             |
| ALPHA     | 2026-01-31 | 2026-02-27 | 2026-02-10 | transactions.amount   | fius91282Y5Y1 |                  | -8.000000        | -8.000000        |                                  | pa: fius91282Y5Y1 transactions.amount decreased by 8.00. This affects the performance calculation through cash-balance ending holdings.market_value.           |
| ALPHA     | 2026-01-31 | 2026-02-27 | 2026-02-17 | transactions.amount   | causCASHUSD   |                  | -2000.000000     | -2000.000000     | 0.001965                         | lo: causCASHUSD transactions.amount decreased by 2,000.00. This affects the performance calculation through weighted external flow, which decreased by 785.71. |
| ALPHA     | 2026-01-31 | 2026-02-27 | 2026-02-27 | holdings.market_value | causCASHUSD   | 7490.100000      | 5482.100000      | -2008.000000     | -0.002018                        | causCASHUSD ending holdings.market_value decreased by 2,008.00.                                                                                                |
| ALPHA     | 2026-01-31 | 2026-02-27 | 2026-02-27 | holdings.quantity     | causCASHUSD   | 7490.100000      | 5482.100000      | -2008.000000     |                                  | causCASHUSD ending holdings.quantity decreased by 2,008.00. This affects the performance calculation through holdings.market_value.                            |
| ALPHA     | 2026-05-01 | 2026-05-29 | 2026-04-30 | holdings.market_value | causCASHUSD   | -350.570000      | -338.480000      | 12.090000        | -0.000012                        | causCASHUSD beginning holdings.market_value increased by 12.09.                                                                                                |
| ALPHA     | 2026-05-01 | 2026-05-29 | 2026-05-29 | holdings.market_value | causCASHUSD   | -4355.530000     | -2855.530000     | 1500.000000      | 0.001426                         | causCASHUSD ending holdings.market_value increased by 1,500.00.                                                                                                |
| ALPHA     | 2026-05-01 | 2026-05-29 | 2026-05-29 | holdings.market_value | csusAAPL      | 226828.750000    | 229097.040000    | 2268.290000      | 0.002157                         | csusAAPL ending holdings.market_value increased by 2,268.29.                                                                                                   |
| ALPHA     | 2026-05-01 | 2026-05-29 | 2026-05-29 | holdings.price        | csusAAPL      | 312.060000       | 315.180600       | 3.120600         |                                  | csusAAPL ending holdings.price increased by 3.12. This affects the performance calculation through holdings.market_value.                                      |
| ALPHA     | 2026-05-01 | 2026-05-29 | 2026-05-29 | holdings.quantity     | causCASHUSD   | -4355.530000     | -2855.530000     | 1500.000000      |                                  | causCASHUSD ending holdings.quantity increased by 1,500.00. This affects the performance calculation through holdings.market_value.                            |
| BALANCED  | 2026-04-01 | 2026-04-10 | 2026-03-31 | holdings.market_value | csusAAPL      | 156324.630000    | 157887.880000    | 1563.250000      | -0.001732                        | csusAAPL beginning holdings.market_value increased by 1,563.25.                                                                                                |
| BALANCED  | 2026-04-01 | 2026-04-10 | 2026-03-31 | holdings.market_value | csusJPM       | 31116.490000     | 31410.650000     | 294.160000       | -0.000326                        | csusJPM beginning holdings.market_value increased by 294.16.                                                                                                   |
| BALANCED  | 2026-04-01 | 2026-04-10 | 2026-04-06 | transactions.amount   | csusJPM       | 148.090000       | 158.670000       | 10.580000        |                                  | dv: csusJPM transactions.amount increased by 10.58. This affects the performance calculation through cash-balance ending holdings.market_value.                |
| BALANCED  | 2026-04-01 | 2026-04-10 | 2026-04-06 | transactions.amount   | csusJPM       | -22.210000       | -23.800000       | -1.590000        |                                  | dp: csusJPM transactions.amount decreased by 1.59. This affects the performance calculation through cash-balance ending holdings.market_value.                 |
| BALANCED  | 2026-04-01 | 2026-04-10 | 2026-04-10 | holdings.market_value | causCASHUSD   | 54138.510000     | 54147.500000     | 8.990000         | 0.000009                         | causCASHUSD ending holdings.market_value increased by 8.99.                                                                                                    |
| BALANCED  | 2026-04-01 | 2026-04-10 | 2026-04-10 | holdings.quantity     | causCASHUSD   | 54138.510000     | 54147.500000     | 8.990000         |                                  | causCASHUSD ending holdings.quantity increased by 8.99. This affects the performance calculation through holdings.market_value.                                |
| BALANCED  | 2026-05-09 | 2026-05-14 | 2026-05-08 | holdings.market_value | csusCVNA      | 293.810000       | 1469.050000      | 1175.240000      | -0.001060                        | csusCVNA beginning holdings.market_value increased by 1,175.24.                                                                                                |



|          |            |            |            |                       |               |               |               |             |           |                                                                                                                                                                                                                   |
|----------|------------|------------|------------|-----------------------|---------------|---------------|---------------|-------------|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| BALANCED | 2026-05-09 | 2026-05-14 | 2026-05-12 | transactions.amount   | csusJPM       |               | 200.000000    | 200.000000  |           | rc: csusJPM transactions.amount increased by 200.00. This affects the performance calculation through cash-balance ending holdings.market_value. Add YAML configuration to count it as explained.                 |
| BALANCED | 2026-05-09 | 2026-05-14 | 2026-05-14 | holdings.market_value | csusMSFT      | 104657.590000 | 105797.530000 | 1139.940000 | 0.001009  | csusMSFT ending holdings.market_value increased by 1,139.94.                                                                                                                                                      |
| BALANCED | 2026-05-09 | 2026-05-14 | 2026-05-14 | holdings.quantity     | csusMSFT      | 255.617800    | 258.402000    | 2.784200    |           | csusMSFT ending holdings.quantity increased by 2.7842. This affects the performance calculation through holdings.market_value.                                                                                    |
| INCOME   | 2026-01-01 | 2026-01-30 | 2026-01-20 | transactions.amount   | causCASHUSD   | -50.000000    | -100.000000   | -50.000000  |           | dp: causCASHUSD transactions.amount decreased by 50.00. This affects the performance calculation through cash-balance ending holdings.market_value.                                                               |
| INCOME   | 2026-01-01 | 2026-01-30 | 2026-01-22 | transactions.amount   | causMARGIN    | -18.750000    | -28.750000    | -10.000000  |           | ai: causMARGIN transactions.amount decreased by 10.00. This affects the performance calculation through cash-balance ending holdings.market_value.                                                                |
| INCOME   | 2026-01-01 | 2026-01-30 | 2026-01-30 | holdings.market_value | causCASHUSD   | 123926.310000 | 123866.310000 | -60.000000  | -0.000060 | causCASHUSD ending holdings.market_value decreased by 60.00.                                                                                                                                                      |
| INCOME   | 2026-01-01 | 2026-01-30 | 2026-01-30 | holdings.quantity     | causCASHUSD   | 123926.310000 | 123866.310000 | -60.000000  |           | causCASHUSD ending holdings.quantity decreased by 60.00. This affects the performance calculation through holdings.market_value.                                                                                  |
| INCOME   | 2026-02-28 | 2026-03-31 | 2026-02-27 | holdings.accrued      | fius91282Y5Y1 | 38.210000     | 38.400000     | 0.190000    | 0.000000  | fius91282Y5Y1 beginning holdings.accrued increased by 0.19.                                                                                                                                                       |
| INCOME   | 2026-02-28 | 2026-03-31 | 2026-02-27 | holdings.market_value | causCASHUSD   | 119921.350000 | 119677.960000 | -243.390000 | 0.000247  | causCASHUSD beginning holdings.market_value decreased by 243.39.                                                                                                                                                  |
| INCOME   | 2026-02-28 | 2026-03-31 | 2026-02-27 | holdings.market_value | fius91282Y5Y1 | 48550.990000  | 48792.430000  | 241.440000  | -0.000245 | fius91282Y5Y1 beginning holdings.market_value increased by 241.44.                                                                                                                                                |
| INCOME   | 2026-04-01 | 2026-04-30 |            |                       |               |               |               |             |           | No identifiable cause was found. Review `source_detail.csv`. The difference may be due to missing source-data, source-file timing differences, or vendor methodology that does not match the YAML specifications. |

## Data Issues

ROWS: 6

| Snapshot   | Portfolio | As Of Date | Dataset.Field       | Security      | Issue Type               | Category         | Reference Value | Observed Value | Difference | Tolerance                   | Explanation                                                                                   |
|------------|-----------|------------|---------------------|---------------|--------------------------|------------------|-----------------|----------------|------------|-----------------------------|-----------------------------------------------------------------------------------------------|
| Snapshot A | ALPHA     | 2026-05-23 | transactions.amount | csusAAPL      | missing_dividend         | income           |                 |                |            | same-date dividend present  | Missing a dividend for csusAAPL on 2026-05-23 that is in portfolio INCOME.                    |
| Snapshot A | INCOME    | 2026-01-30 | holdings.price      | csusGOOGL     | holdings_price_range     | price            | 313.000000      | 338.000000     | 25.000000  | greater of 0.01 or 0.05%    | Same-day same-security holdings.price differs across portfolios for csusGOOGL.                |
| Snapshot B | ALPHA     | 2026-02-10 | transactions.amount | fius91282Y5Y1 | pa_sa_rate               | accrued_interest | 0.104377        | 0.159133       | 0.054757   | greater of 0.000001 or 0.5% | pa: Same-day same-security rate differs across portfolios for fius91282Y5Y1.                  |
| Snapshot B | BALANCED  | 2026-04-06 | transactions.amount | csusJPM       | dividend_rate            | income           | 1.399942        | 1.500012       | 0.100069   | greater of 0.01 or 0.5%     | dv: Same-day same-security rate differs across portfolios for csusJPM.                        |
| Snapshot B | INCOME    | 2026-01-30 | holdings.accrued    | fius91282Y5Y1 | holdings_accrued_rate    | accrued_interest | 0.094983        | 0.096077       | 0.001094   | greater of 0.00001 or 0.5%  | Same-day same-security holdings.accrued per unit differs across portfolios for fius91282Y5Y1. |
| Snapshot   | INCOME    | 2026-03-05 | transactions.price  | csusAAPL      | transactions_price_range | price            | 260.290000      | 260.440000     | 0.150000   | greater of                  | Same-day same-security                                                                        |

|   |  |  |  |  |  |  |  |  |  |  |               |                                                            |
|---|--|--|--|--|--|--|--|--|--|--|---------------|------------------------------------------------------------|
| B |  |  |  |  |  |  |  |  |  |  | 0.01 or 0.05% | transactions.price differs across portfolios for csusAAPL. |
|---|--|--|--|--|--|--|--|--|--|--|---------------|------------------------------------------------------------|

## Setup

PPAR requires Python 3.11.9 or newer.

Install the PPAR package:

```
pip install ppar
```

Create a local PPAR Audit workspace. The workspace includes PPAR-normalized demonstration data modeled on Axys/APX source and report data, so you can run the complete workflow before replacing the CSV files with reviewed exports from your own environment.

```
ppar setup ./my_ppar_audit
```

Run Audit:

```
ppar audit ./my_ppar_audit
```

Follow the `Customizing With Your Own Data` section in `./my_ppar_audit/README.md` when you are ready to customize the workspace with your own data.

```
my_ppar_audit/  
  README.md  
  ppar.yaml  
  run_audit.py  
  snapshot_a/  
    portperf.csv  
    holdings.csv
```

```
transactions.csv
secmast.csv
secperf.csv
splits.csv
snapshot_b/
  portperf.csv
  holdings.csv
transactions.csv
secmast.csv
secperf.csv
splits.csv
```

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## Inputs

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The demonstration files are PPAR-normalized CSVs modeled on Axys/APX data. Production inputs may come from reviewed REP, IMEX, custom-report, or other controlled exports:

- portfolio performance;
- security performance;
- holdings;
- transactions;
- security master data;
- split factors.

Audit uses two source-data snapshots. Snapshot A is normally the older or original state and Snapshot B is normally the newer or restated state, but neither snapshot is presumed correct.

PPAR normalizes those files through a customizable `ppar.yaml` file, so each site can configure its local field names, transaction-code treatment, comparison tolerances, and report assumptions. The setup-created file documents these choices in place.

The configured file and accounting contracts fail closed when required source treatment is missing or ambiguous. Optional evidence does not silently expand the calculation or policy surface.

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# Outputs

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PPAR Audit writes review packages:

```
output/  
  portfolio/  
    portfolio_audit.xlsx  
    portfolio_audit.html  
    source_detail.csv  
    audit_support.zip  
  security/  
    security_audit.xlsx  
    security_audit.html  
    source_detail.csv  
    audit_support.zip
```

To prevent unusably large artifacts, Audit stops with a nonzero exit code before writing a report when any primary review table would exceed 100,000 rows. The error identifies the oversized table and its largest contributors so the user can narrow the portfolio or date scope or correct upstream differences.

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## Current Validation Scope

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PPAR Audit has substantial automated coverage, financial invariants, report reconciliation checks, output-integrity checks, deterministic demonstrations, and maintained scale gates.

It has not yet been validated against a real client's production-style Axys/APX exports and approved local accounting policy. The current program is seeking a small number of strong validation partners to test source authenticity, setup burden, financial interpretation, false positives, and reviewer usefulness.

PPAR Audit detects, compares, explains, and helps investigate supported portfolio-performance and source-data differences. It does not provide a financial-statement audit, GIPS verification, attestation, certification, or assurance opinion.

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## Additional Repository Capability

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This repository also contains `ppar.analytics`, a maintained module for benchmark-relative performance attribution, contribution, and ex-post risk reporting. It is retained for future PPAR packaging but is not part of the current PPAR Audit validation program or default onboarding workflow.